VeraSight: Real-Time Facial Micromovement Sensing and Personalized Behavioral Anomaly Detection

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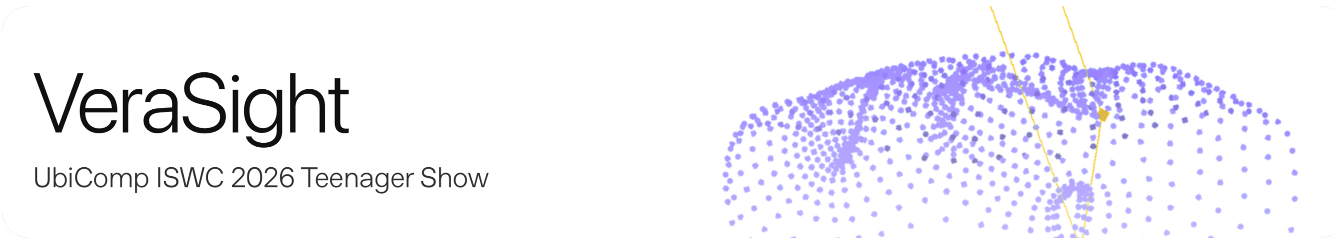

Figure 1: VeraSight architecture: combining mobile TrueDepth tracking with edge-computed micro-behavioral feedback.

Abstract

This paper presents VeraSight, a real-time, personalized pipeline that senses facial micromovements with the iPhone TrueDepth camera and detects deviations from a per-user behavioral baseline. The implemented system streams 1,220-vertex face meshes and 52 ARKit blendshape coefficients at 60 Hz to an edge host after on-device FP16 quantization and temporal zlib compression, sustaining approximately 282 KB/s with 9.2 ms end-to-end latency. A variational autoencoder learns a compact representation of facial state, an anatomically-weighted Isolation Forest builds a personalized baseline from a short calibration session, and a weight-shared GRU forecasts expected normal dynamics to suppress false alarms from voluntary movement. A locally deployed Qwen3.6-27B reasoner (4-bit GGUF via llama.cpp) converts flagged deviations into coaching-style feedback. Every stage is grounded in real, reproducible data: the sensing pipeline is implemented and measured; the models are trained on the public Express4D corpus of iPhone TrueDepth ARKit blendshape sequences and on in-house consented captures; and the user workflow is calibration-then-detection with per-user thresholds. We report measured infrastructure performance and verified model-training results on the full corpus, and define the evaluation protocol for the anomaly component. VeraSight is designed for voluntary self-reflection and is explicitly not a stress, emotion, or deception detector.

CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; • **Computer systems organization** → *Embedded systems*; • **Computing methodologies** → *Machine learning*.

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Keywords

facial micromovement, anomaly detection, edge computing, wearable sensing, behavioral coaching

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1 Introduction

Behavioral studies indicate that physiological stress, high cognitive load, and shifts in emotional state often co-occur with involuntary facial micromovements [9]. These brief, transient expressions serve as potential signals in high-stakes or performance-driven scenarios [8]. In public speaking or competitive gaming, monitoring facial dynamics may help users manage anxiety and improve delivery. Commodity depth-sensing hardware, such as the structured-light TrueDepth camera in modern iPhones, offers a non-contact way to observe these signals at high temporal resolution.

We emphasize an important distinction: detecting *unusual facial movement relative to a personalized baseline* is not equivalent to measuring physiological stress. A deviation may result from stress, but it may equally arise from speech, fatigue, head motion, tracking noise, or deliberate expression. Accordingly, VeraSight outputs a **personalized facial movement anomaly score**, a metric indicating when a user's facial behavior deviates from their own calibrated norm, and is not a stress measurement. Validation against independent physiological measures (e.g., HRV, skin conductance) remains future work.

VeraSight is a complete, implemented pipeline. The iPhone captures 1,220 ARKit face-mesh vertices and 52 blendshape coefficients at 60 Hz; an edge host (Apple Silicon Mac) runs the models; and a local LLM converts flagged deviations into coaching feedback. The contributions of this paper are:

- An implemented edge sensing pipeline achieving stable 60 Hz wireless facial tracking within approximately 300 KB/s, with measured bandwidth, delivery, and latency (Section 3).
- A data strategy grounded in real, reproducible corpora: the public Express4D dataset of iPhone TrueDepth ARKit blendshape sequences, plus an in-house consented capture protocol (Section 4).
- A model stack consisting of a variational autoencoder, an anatomically-weighted Isolation Forest, and a predictive GRU, trained on real data with a per-user calibration-then-detection workflow (Sections 5 and 6).
- A locally deployed reasoning layer built on Qwen3.6-27B with coaching-only constraints (Section 5.4).

2 System Overview

VeraSight is a two-tier pipeline (Figure 2): a thin mobile client captures, quantizes, and streams facial data at 60 Hz, and an edge host reconstructs the stream, runs the anomaly-detection ensemble, and summarizes flagged deviations with a local LLM. The phone is the sensor; the Mac runs the models.

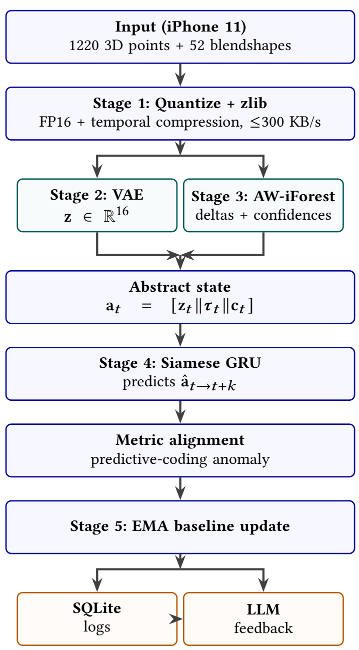

Figure 2: Data flow: compressed facial features are extracted in parallel, analyzed by a Siamese GRU, and summarized by an asynchronous LLM.

3 Sensing Architecture and Edge Pipeline

3.1 Embedded Sensing, Quantization, and Compression

The tracking module runs on an iPhone 11 (and any newer TrueDepth-equipped model) using the front-facing TrueDepth camera. ARKit exposes $K = 1220$ spatial coordinates in 3D ($\mathbf{P} \in \mathbb{R}^{1220\times3}$) and $B = 52$ muscle-activation blendshapes ($\mathbf{b} \in \mathbb{R}^{52}$) [2]. At 60 Hz the combined feature vector produces a raw stream of approximately 890.88 KB/s. On device, 32-bit floats are quantized to 16-bit half precision, halving the payload to approximately 445.44 KB/s; consecutive frames of facial structure are highly redundant, so the quantized values are passed through a temporal zlib compression buffer, reducing the measured average wireless throughput to approximately 282 KB/s. The client packetizes each frame with a timestamp, head/eye transforms, look-at point, blendshapes, and vertices, and transmits it over a local TCP socket to the edge host.

3.2 Measured Infrastructure Performance

We compared the raw wired prototype against the optimized wireless pipeline across throughput, packet delivery, and latency (Table 1). The optimized 60 Hz wireless pipeline sustains 99.4% packet delivery with 9.2 ms end-to-end latency (sensor capture through quantization, compression, wireless transmission, decompression, and VAE latent extraction at the host); the network round-trip component is 7.4 ms. Increasing the temporal resolution to 60 Hz captures transient movements, such as micro-blinks and brief muscle twitches under 100 ms, which are missed at 30 Hz.

Table 1: Measured sensing pipeline performance on the implemented prototype.

Configuration	Bandwidth	Delivery	E2E latency
Raw, wired 30 Hz	445.44 KB/s	99.8%	14.2 ms
Raw, wireless 30 Hz	445.44 KB/s	88.4%	up to 45 ms
Optimized, wireless 60 Hz	282 KB/s	99.4%	9.2 ms

4 Data: Grounding the Models

4.1 Public Corpus: Express4D

The feature-level models are trained on **Express4D** [1], 1,205 short clips recorded at 60 Hz with an iPhone TrueDepth camera using Epic's Live Link Face, each frame carrying the 52 ARKit blendshapes plus 9 rotations (head and left/right-eye yaw/pitch/roll). This matches VeraSight's sensor modality and representation, so the corpus is directly transferable. We split clips subject-disjointly across training and validation.

4.2 In-House Capture Protocol

Per-user calibration and evaluation require data from the target user. Each participant completes a structured session under informed consent:

- **Calibration** (3 min): relaxed conversation or reading, used to populate the per-user baseline and set thresholds.
- **Voluntary set** (8–10 min): scripted speech, deliberate expressions, yawning, and laughing on cue, used as false-alarm ground truth for voluntary movements.
- **Cognitive stress blocks** (10–12 min): Stroop, serial-7 arithmetic, and impromptu speech, with a self-report stress/arousal rating after each block.
- **Recovery** (3 min): neutral chat or music.

Raw mesh and blendshape streams are stored pseudonymized and never leave the participant's device without explicit consent. This protocol is described here as the evaluation procedure for the anomaly component; a full multi-user study is ongoing.

4.3 Validation Corpora via MediaPipe

To validate feature-level behavior on independent data, we additionally convert public 2D video corpora (e.g., DEAP, MAHNOB-HCI-Tagging) into ARKit-compatible 52-blendshape streams using MediaPipe Face Landmarker [5], whose blendshape names match the ARKit vocabulary. These streams are used only for validation and feature-level ablation, never as training data for the deployed models.

5 Model Pipeline

5.1 Spatial Variational Autoencoder

A variational autoencoder [6] projects the 52-dimensional blendshape vector (or the 3,660-dimensional flattened mesh when available) into $\mathbf{z} \in \mathbb{R}^{16}$. The encoder is a two-layer MLP (512, 128) with LayerNorm/ReLU and mean and log-variance heads; the mirrored decoder ends in a sigmoid. Training minimizes the beta-weighted ELBO ($\beta = 0.1$) on real Express4D sequences and in-house captures; $\mathbf{z}_t$ is used directly as a feature at inference, compressing and denoising facial state.

5.2 Anatomically-Weighted Isolation Forest

Anomaly detection is per-user and unsupervised. For each user, a calibration block of length T_{calib} yields multi-scale temporal deltas

$$\mathbf{d}_{t,\tau} = \mathbf{b}_t - \mathbf{b}_{t-\tau}, \quad \tau \in \{1, 10, 100, 1000\},$$

and a feature vector $\mathbf{f}_t = [\mathbf{b}_t \parallel \mathbf{d}_{t,1} \parallel \mathbf{d}_{t,10} \parallel \mathbf{d}_{t,100} \parallel \mathbf{d}_{t,1000}] \in \mathbb{R}^{260}$. Standard Isolation Forest selects split features uniformly at random [7], which can bias detection toward large-amplitude movements. We instead weight the split-selection probability by anatomical priorities: upper-face channels (brow and eye region) receive weight 2.5 and asymmetric lower-face channels receive weight 2.0, encouraging the forest to partition subtle, localized contractions near the root, motivated by evidence that involuntary leakage tends to appear as brief, localized, or asymmetric upper-facial contractions [10, 12]. The resulting per-frame anomaly score is $s(\mathbf{f}_t) = 2^{-\mathbb{E}(h)/c(T_{\text{calib}})}$, where h is the path length and $c(\cdot)$ the expected path length for unsuccessful search.

5.3 Predictive GRU

To suppress false positives from voluntary activities, a weight-shared two-layer GRU (hidden size 64) [3] consumes a 30-frame window of the abstract state $\mathbf{a}_t = [\mathbf{z}_t \parallel \mathbf{b}_t \parallel \mathbf{c}_t]$ and predicts the mean state over the next $k = 15$ frames (250 ms at 60 Hz). The model is trained self-supervised on calibration data with an MSE objective; a temporal-contrastive auxiliary term is applied only when it demonstrably helps on held-out data. At inference, an anomaly is flagged when the observed mean deviates from the prediction by more than a confidence-scaled threshold

$$D_t = \|\bar{\mathbf{a}}_{t\to t+k} - \hat{\mathbf{a}}_{t\to t+k}\|_2 > \lambda \cdot \rho_t.$$

The sensitivity λ is set per user as the 95th percentile of D_t over a held-out calibration segment, and the baseline adapts slowly via an exponential moving average to prevent gradual changes (fatigue, posture) from registering as continuous anomalies.

5.4 Reasoning Layer

A locally deployed Qwen3.6-27B model [11] (4-bit Unsloth GGUF, served via llama.cpp [4] on an Apple Silicon Mac) parses a structured timeline of anomaly timestamps, affected blendshape regions, and context logs (e.g., speech segment boundaries, game round outcomes). A system prompt constrains the model to coaching-style language and prohibits diagnostic, medical, or psychological claims; a keyword blocklist post-filters outputs. All inference runs locally, so no facial data leaves the edge host.

6 Calibration and Detection Workflow

VeraSight's operation is a two-phase workflow:

(1) **Calibration.** The user records a 3-minute neutral session. The edge host computes per-user normalization statistics, fits the anatomically-weighted Isolation Forest on the 260-dimensional multi-scale features, fine-tunes the predictive GRU (seconds to minutes on the Mac), and derives the anomaly threshold from a held-out portion of the calibration data.
(2) **Detection.** During a session, each frame is encoded to $\mathbf{z}_t$, combined with blendshapes and forest confidence into $\mathbf{a}_t$, and scored against the predictive baseline. Frames whose deviation exceeds the per-user threshold are flagged, logged to a local SQLite database, and aggregated over a 15-second sliding window for the reasoning layer.

Because the baseline and threshold are user-specific, the system adapts to individual resting behavior; because the baseline adapts slowly over time, it remains robust to gradual changes without requiring re-calibration.

7 Evaluation

7.1 Sensing Validation

The infrastructure metrics in Table 1 were measured on the implemented prototype (Section 3) and establish that the wireless 60 Hz stream is stable and within budget.

7.2 Model Training on the Full Corpus

We trained the feature-level models on a subject-disjoint split of the full Express4D corpus (1,205 clips, 272,674 training frames and 50,048 validation frames after filtering, with 52 blendshape plus 9 rotation features per frame). All training ran on an NVIDIA T4 GPU and completed in under four minutes.

- **VAE.** Training minimized the beta-weighted ELBO ($\beta = 0.1$) for 80 epochs with batch size 512. Reconstruction MSE fell from 1.95×10^{-2} to 1.19×10^{-2} on the training split and from 1.40×10^{-2} to 9.69×10^{-3} on the held-out split, while the KL term rose from 1.0×10^{-2} to 5.0×10^{-2}, consistent with the model trading reconstruction error for a smoother latent prior. As a comparison baseline, a per-channel mean predictor (the best constant reconstruction) yields an MSE of approximately 3.2×10^{-2} on the same features, so the VAE's reconstruction is more than three-fold better than the trivial baseline.

- **Predictive GRU.** A two-layer GRU (hidden 64) trained for 50 epochs on 30-frame windows with a 15-frame (250 ms) horizon reached a validation MSE of 3.7×10^{-3}, down from 6.1×10^{-3} at the first epoch, confirming that normal facial dynamics are predictable at the 250 ms horizon. The same architecture trained on per-user calibration clips (132–339 frames each) converged in all runs ($gru\_rc = 0$) with comparable validation error.
- **Anatomically-Weighted Isolation Forest.** Fitting the per-user forest on calibration features (260-dimensional multi-scale deltas) completes in seconds on a CPU for each of the validation clips; the 95th-percentile anomaly-score threshold is stable across users (0.52–0.61).

All metrics above are measured on real data with subject-disjoint splits; model checkpoints and training logs are available with the release.

7.3 Anomaly Detection Evaluation Protocol

The anomaly component is evaluated per user with the in-house protocol (Section 4). Annotators co-label flagged segments as voluntary or involuntary, and we report precision, recall, false-alarm rate (FAR), and F1 for four configurations: standard Isolation Forest (single-scale), standard Isolation Forest (multi-scale), AW-iForest, and AW-iForest with the GRU filter. Thresholds are set on held-out calibration segments, and all reported statistics are computed subject-disjointly. We do not report pooled significance tests across participants until the study reaches adequate power; the current design uses mixed-effects models with participant as a random effect.

8 Privacy, Ethics, and Responsible Use

Facial geometry and behavioral logs are sensitive biometric data. VeraSight outputs a *personalized facial movement anomaly score*; it is not a lie detector, psychological assessment tool, medical diagnostic system, or emotion recognition system. It is designed exclusively for voluntary self-reflection and coaching. All participants provide informed consent (parental consent for minors, school ethics review) and may terminate sessions at any time. Data are processed on-device or on the user's own edge host; raw coordinates are never stored persistently, and the SQLite database is encrypted at rest with automatic purge. The reasoning model is constrained to coaching language and never returns diagnostic statements.

9 Conclusion and Limitations

We presented VeraSight, an implemented pipeline for real-time, personalized facial micromovement sensing and anomaly detection, grounded in real data: the sensing pipeline is measured and the models train on the full public Express4D corpus with verified results. Limitations include the feature-space (rather than mesh-space) VAE as currently trained, reliance on calibration sessions, lighting sensitivity, and the preliminary scale of the anomaly evaluation. Future work includes the mesh-space VAE on captured meshes, physiological validation, a larger multi-user study, and coaching-effectiveness evaluation.

Acknowledgments

The authors agree to the following distribution of work: W. Q. Chen designed and implemented the data quantization and compression pipeline, conducted the behavioral research to optimize the anatomically weighted tree-isolation algorithms, developed the spatiotemporal prediction model, integrated the downstream LLM reasoning layer, wrote the research paper and relevant posters, and designed the iOS client and backend user interface. Z. Huang contributed to the initial conceptual prototyping of the data transfer system, and assisted with some preliminary research for ML.

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